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$$z^2 + 2iz - 9 - 6i = 0$$

$$D = (2i)^2 - 4 \cdot (-9 - 6i) = -4 + 36 + 24i = 32 + 24i$$

$$\lambda^2 - 32\lambda - 144 = 0$$

$$D = (-32)^2 - 4 \cdot (-144) = 1600$$

$$\lambda_1 = \frac{32 + 40}{2} = 36 \Rightarrow x = \pm 6$$

$$\lambda_2 = \frac{32 - 40}{2} = -4 \Rightarrow y = \pm 2$$

$$z_1 = \frac{-2i + 6 + 2i}{2} = 3$$

$$z_2 = \frac{-2i - 6 - 2i}{2} = -3 - 2i$$

References